

MICROSOFT SENTINEL-STYLE

SIEM Incident Report

Suspicious Authentication, Account Compromise, and External Email Disclosure

Organization	Northstar Financial Services (fictional)
Affected Account	mlopez@northstarfinancial.com
Incident Date	October 1, 2026
Final Severity	High
Status	Contained - Further impact assessment recommended
Environment	Synthetic Microsoft Sentinel-style training lab

Portfolio Training Disclaimer

This report documents a simulated cybersecurity investigation using synthetic telemetry. No production systems, real customer data, or live Microsoft Sentinel tenant were used.

1. Executive Summary

Microsoft Sentinel generated a Medium-severity alert after 11 failed authentication attempts against the M. Lopez account were followed by a successful authentication from the same source IP. Investigation found that the successful authentication originated from a previously unobserved IP address in Amsterdam and an unknown, unmanaged Linux device inconsistent with the user's recent baseline.

Post-authentication review identified unauthorized Microsoft 365 mailbox activity, downloads of business and payroll information, SharePoint access to Finance/Accounts Payable data, and download of a document containing full vendor routing and bank-account information. An unauthorized external mailbox forwarding rule resulted in six organizational emails being forwarded outside Northstar, including an accounts-payable approval request and a vendor bank-change request.

The incident was escalated to High severity. Security disabled the affected account, revoked active sessions, reset credentials, preserved forwarding-rule evidence, and removed the unauthorized rule. Available evidence confirmed external disclosure of email data but did not establish that a fraudulent payment occurred or that the downloaded SharePoint data was transferred through an additional exfiltration mechanism.

2. Incident Classification

Initial Alert Severity	Medium
Final Incident Severity	High
Incident Type	Unauthorized account access / cloud account compromise
Primary Systems	Microsoft 365, Exchange Online, SharePoint
Affected User	mlopez@northstarfinancial.com
Suspicious Source IP	45.83.64.19
Suspicious Location	Amsterdam, Netherlands
Suspicious Device	Unknown, unmanaged Linux device

3. Initial Alert and Authentication Analysis

The alert was treated as an investigative lead rather than automatic proof of compromise. The investigation first established the user's recent authentication baseline and compared the alert activity against normal source IP, location, and device behavior.

Recent Baseline

- Source IP: 10.24.18.56
- Location: Orlando, US
- Device: NFS-LT-0442
- Operating system: Windows 11
- Managed device: Yes

Suspicious Authentication

- 11 failed sign-ins between approximately 8:31 AM and 8:43 AM
- Successful authentication at approximately 8:44 AM
- Source IP: 45.83.64.19
- Location: Amsterdam, Netherlands

- Device: Unknown
- Operating system: Linux
- Managed device: No

The suspicious IP was not observed with other Northstar users in the available synthetic telemetry. No single anomaly was treated as conclusive; the combined pattern of repeated failures, subsequent success, new source IP, unusual location, and unmanaged device supported an assessment of likely unauthorized access.

4. Incident Timeline

Time	Observed Activity
08:31-08:43	11 failed authentication attempts from 45.83.64.19
08:44	Successful authentication from the same source IP
08:46	Exchange Online accessed
08:48	Mailbox search performed for "invoice"
08:50	Email "Q3 Vendor Payment Schedule" opened
08:52	Vendor_Payment_Schedule_Q3.xlsx downloaded
08:55	Email "October Payroll Processing" opened
08:57	October_Payroll.xlsx downloaded
09:01	SharePoint accessed
09:03	Finance/Accounts Payable folder accessed
09:06	Vendor_Banking_Information.xlsx downloaded
09:09	Unauthorized external mailbox forwarding rule created
09:13	Sixth confirmed external email forward occurred

5. Post-Compromise Activity and Data Impact

Vendor_Payment_Schedule_Q3.xlsx

Contained vendor names, invoice numbers, payment amounts, scheduled payment dates, and internal accounts-payable notes. No personal information or bank-account information was identified, but the file contained sensitive internal business and financial information.

October_Payroll.xlsx

Contained employee names, employee ID numbers, departments, gross salary, net pay, and the last four digits of employee bank accounts. The file contained employee PII and sensitive payroll information; Social Security numbers and full employee bank-account numbers were not present.

Vendor_Banking_Information.xlsx

Contained vendor company names, bank names, full routing numbers, full bank-account numbers, accounts-payable contact names, and contact email addresses. This represented the most significant financial exposure identified during the investigation.

6. External Forwarding and Fraud Risk

The unauthorized forwarding rule directed incoming email to payments.review@external-mail.example. Six messages were confirmed as forwarded externally before containment:

- Vendor invoice notification
- Internal meeting reminder
- Vendor payment confirmation
- Employee benefits announcement
- Accounts-payable approval request
- Vendor bank-change request / updated banking instructions

The forwarding activity confirms external disclosure of organizational email data. The combination of vendor banking information, payment-process visibility, an accounts-payable approval request, and a vendor bank-change request created significant risk of business email compromise and payment fraud. No evidence in the available dataset confirmed that banking information was modified or that a fraudulent payment occurred.

7. Severity Assessment

The incident was escalated from Medium to High severity based on confirmed unauthorized account access, access to sensitive Microsoft 365 and SharePoint resources, download of employee and vendor financial information, exposure of full vendor routing and bank-account numbers, creation of an unauthorized forwarding rule, and confirmed forwarding of six organizational emails to an external destination.

8. Containment and Response

- Disabled the compromised M. Lopez account.
- Revoked active sessions and authentication access.
- Initiated a credential reset.
- Reviewed account authentication and MFA configuration.
- Preserved details of the unauthorized forwarding rule as evidence.
- Removed the unauthorized forwarding rule.
- Preserved relevant authentication, Microsoft 365, SharePoint, and forwarding activity for further investigation.

9. Stakeholder Escalation

Because the incident involved employee information, vendor banking information, confirmed external email forwarding, and potential payment-fraud risk, the incident was escalated beyond the SOC.

- SOC and security leadership
- Finance and Accounts Payable
- Legal, privacy, and compliance personnel as appropriate
- Incident response stakeholders
- Relevant business and data owners

Finance and Accounts Payable should review recent and pending vendor banking changes and payment activity for unauthorized modification or fraudulent transactions. Any external vendor notification should be coordinated according to organizational incident-response procedures.

10. MITRE ATT&CK Mapping

Technique	Observed Behavior	Assessment
T1110 - Brute Force	Repeated password attempts preceded the successful authentication.	Consistent with brute-force behavior in this simulated scenario.
T1114 - Email Collection	Mailbox searched; financial emails opened; attachments downloaded.	Confirmed collection behavior.
T1114.003 - Email Forwarding Rule	Unauthorized rule forwarded incoming mail to an external address.	Six messages confirmed externally forwarded.

The SharePoint download confirms unauthorized collection of sensitive financial information. The available evidence does not establish whether that downloaded file was subsequently transferred through an additional exfiltration mechanism. External forwarding of six mailbox messages separately establishes confirmed external disclosure.

11. Detection Engineering

A Microsoft Sentinel-style KQL detection was developed to identify similar authentication behavior. The training detection looks for five or more failed sign-ins for the same user and source IP within a 15-minute window, followed by a successful authentication, with an unmanaged device or unusual location as contextual enrichment.

- Threshold: 5 or more failed sign-ins
- Window: 15 minutes
- Sequence: successful authentication must occur after the failed attempts
- Context: unmanaged device OR unusual location
- False-positive considerations: legitimate travel, VPN use, password-entry mistakes, authorized temporary devices, and known network ranges

The hard-coded Orlando location used in the lab is intentionally simplified. A production implementation should use historical user baselines, approved VPN/corporate ranges, asset inventory, travel context, and other enrichment rather than treating one city as universally normal.

12. Recommendations

- Review recent and pending vendor payment and banking changes for unauthorized modification.
- Validate affected vendors and financial workflows through approved Finance/Accounts Payable procedures.
- Review mailbox, SharePoint, and identity telemetry for additional unauthorized activity.
- Assess whether additional employee or vendor information was exposed.
- Review MFA methods, token/session activity, and identity controls for persistence or abuse.
- Tune and test the failed-then-success authentication detection against legitimate activity.
- Enrich future alerts with device inventory, corporate/VPN ranges, historical sign-in baselines, and approved travel context.
- Maintain separate detections for high-volume failed authentication activity that does not result in successful access.

13. Final Analyst Conclusion

The investigation identified a compromised Microsoft 365 account following repeated failed authentication attempts and a subsequent successful login from a previously unobserved source IP, unusual location, and unknown unmanaged device. The unauthorized session accessed and downloaded sensitive business, payroll, and vendor banking information, created an external mailbox forwarding rule, and resulted in six organizational emails being forwarded outside Northstar.

Security contained the account and removed the forwarding mechanism while preserving relevant evidence. The incident was classified as High severity and escalated to appropriate security, Finance, Legal/Compliance, incident response, and business stakeholders. Confirmed external disclosure was limited by the available evidence to the six forwarded emails; additional exfiltration through other mechanisms was not established in the synthetic dataset.

14. Skills Demonstrated

- SIEM alert triage and investigation
- Authentication baseline analysis
- KQL fundamentals and query development
- Event correlation and scope analysis
- Microsoft 365 and SharePoint activity analysis
- Detection engineering and threshold design
- False-positive analysis and alert enrichment
- MITRE ATT&CK mapping
- Incident severity assessment
- Containment and stakeholder escalation
- Evidence-based technical documentation